Lecture-15

Vt standard

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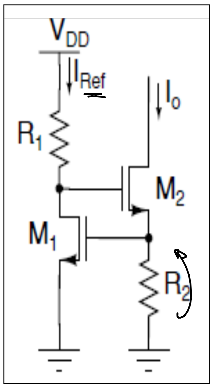


$$I_o = \frac{V_{GS1}}{R_2} = \frac{V_t + V_{OV1}}{R_2} = \frac{\cancel{V_t} + \sqrt{\frac{2I_{Ref}}{\underbrace{\mu C_{ox}\left(\frac{W}{L}\right)_1}_{\beta_1}}}}{R_2}$$

$$\frac{\partial I_o}{\partial I_{REF}} = \frac{1}{R_2}\cdot\sqrt{\frac{2}{\beta_1}}\cdot\frac{1}{2}\cdot\frac{1}{(\sqrt{I_{REF}})} = \frac{1}{R_2\,\beta_1\,V_{OV1}}$$

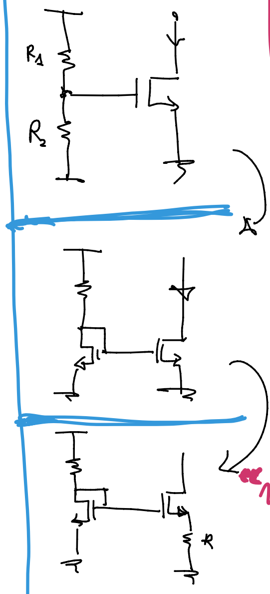
$$\sqrt{\frac{2I_{REF}}{\beta_1}} = V_{OV1} \Rightarrow \sqrt{I_{REF}} = \sqrt{\frac{\beta_1}{2}}\cdot V_{OV1}$$

$$S^{I_o}_{I_{REF}} = \frac{\partial I_o}{\partial I_{REF}}\cdot\frac{I_{REF}}{I_o} = \frac{1}{\cancel{R_2}\beta_1 \cancel{V_{OV1}}}\cdot\frac{\frac{\cancel{\beta_1}}{2}\cdot V_{OV1}^{\cancel{2}}}{\frac{V_T + V_{OV1}}{\cancel{R_2}}}$$

$$\boxed{S^{I_o}_{V_{DD}} = S^{I_o}_{I_{REF}}\cdot S^{I_{REF}}_{V_{DD}}}$$

$$\boxed{S^{I_o}_{I_{REF}} = \frac{1}{2}\cdot\frac{V_{OV1}}{V_T + V_{OV1}}}$$

$$\boxed{S^{I_o}_{V_{DD}} = \frac{\partial I_o}{\partial V_{DD}}\cdot\frac{V_{DD}}{I_o}}$$



$S\downarrow$

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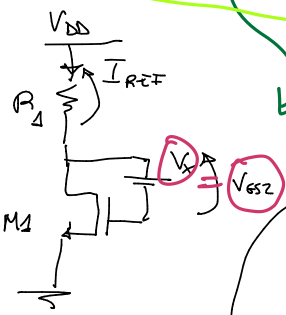
Sensitivity is Decreasing

"widlar source"



$$I_{REF} = \frac{V_{DD} - V_{OV1} - V_T - V_x}{R_1}$$

$$S^{I_{REF}}_{V_{DD}} = \frac{\partial I_{REF}}{\partial V_{DD}}\cdot\frac{V_{DD}}{I_{REF}} \sim 1$$

$$\frac{\partial I_{REF}}{\partial V_{DD}} \simeq \frac{1}{R_1}$$

$$V_{DD} = I_{REF}\cdot R_1 + (\,\dots\,)$$

$$\frac{V_{DD}}{I_{REF}} = R_1 + \left(\frac{\dots}{I_{REF}}\right) \simeq R_1$$

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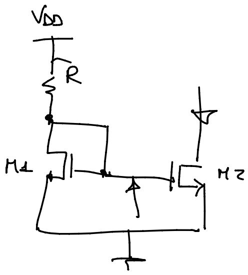

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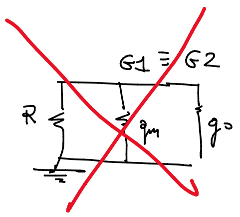

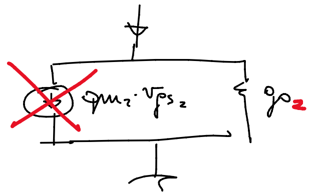

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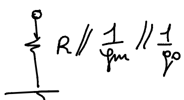
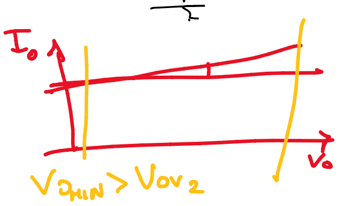
i_{o2}

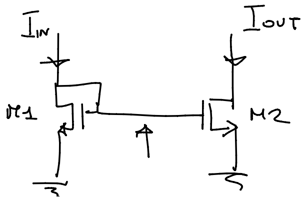


$v_{gs2} = 0$



$V_{o,MIN} > V_{OV2}$

$$R_{in} = \frac{1}{g_m + g_o} \checkmark$$



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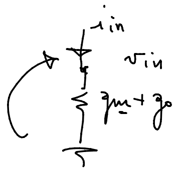
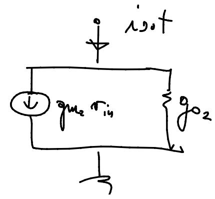
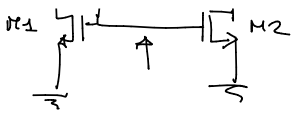
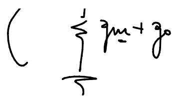


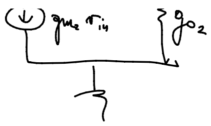


$$R_o = \frac{1}{g_{o2}} \checkmark$$

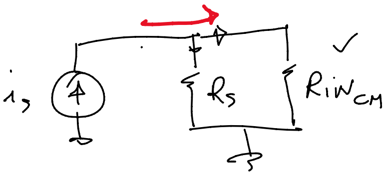


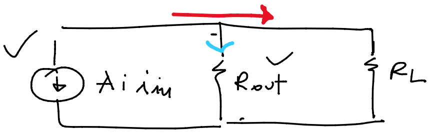
$\longrightarrow$





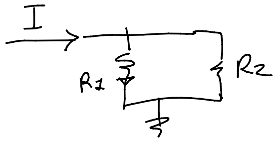
$R_o = \frac{1}{g_{o2}}$ ✓





$v_{in} \sim \frac{i_i}{g_m}$

$g_{m2}\, v_{in} = \frac{g_{m2}}{g_{m1}} \cdot i_i$



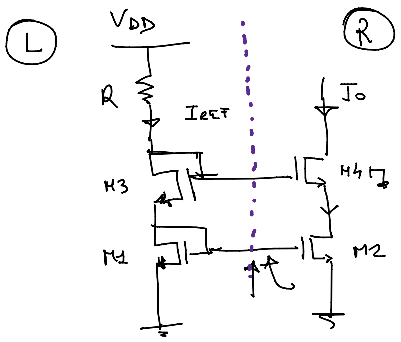
$A_i = \frac{g_{m2}}{g_{m1}} \sim 1$

$I_{R1} = I \, \frac{R_2}{R_1 + R_2}$

$I_{R2} = I \cdot \frac{R_1}{R_1 + R_2}$

$A_i = \frac{i_o}{i_i}\Big|_{v_o = 0}$

$R_1 \gg R_2$

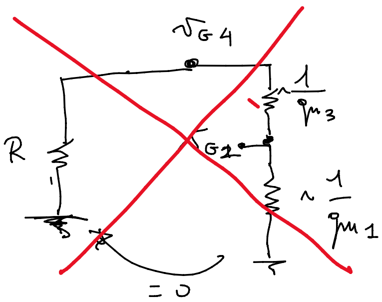


$I_{REF} = I_3 = I_1$

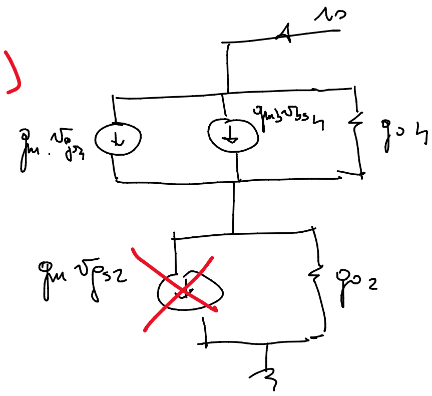
$I_o = I_4 = I_2$

If M2 is SAT
$\frac{W}{L}_1 = \frac{W}{L}_2$ $\Big\}$ $\rightarrow I_2 \sim I_1 \Rightarrow I_o \sim I_{REF}$

S.S.C (L)



(R)

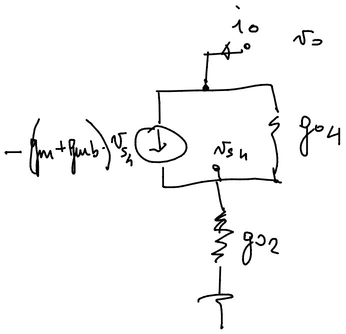


$v_{gs4} = \cancel{v_{g4}} - v_{s4}$

$v_{gs4} = -v_{s4}$

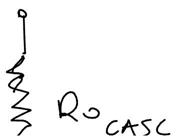
$v_{bs4} = -v_{s4}$

$v_{G4} = v_{G1} = 0$



$\leadsto R_{o_{CASC}} = \frac{v_o}{i_o} = \frac{G_{T4}}{g_{o2} \cdot g_{o4}} \sim g_{m4} \cdot r_{o4} \cdot r_{o2}$

S.S.C



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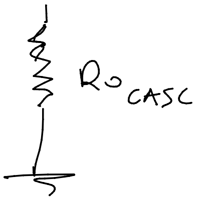
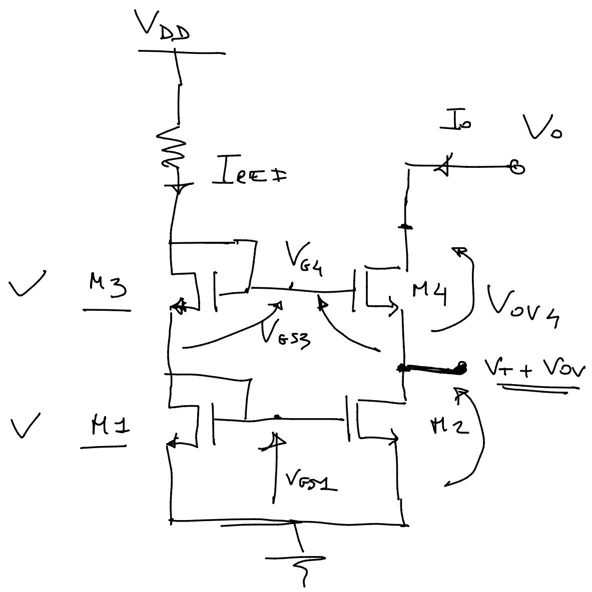




$$\boxed{V_{OMIN}}$$

$$\cancel{V_{DS}} = V_{GS} - V_T$$

$$V_{DS} = V_{GS}$$

Wrong Answer

$$[\, V_{DS_2} > V_{ov} \,]$$

$$V_{DS_2} = V_{ov_2}$$

$$\cancel{V_{OMIN} = V_{ov_2} + V_{ov_4}}$$

$$V_{GS1} = \underbrace{V_T + V_{ov_1}}$$

$$V_{GS3} = V_T + V_{ov_3}$$

$$\frac{W}{L_1} = \frac{W}{L_3} \Rightarrow V_{ov_1} = V_{ov_3}$$

$$I_1 = I_3$$

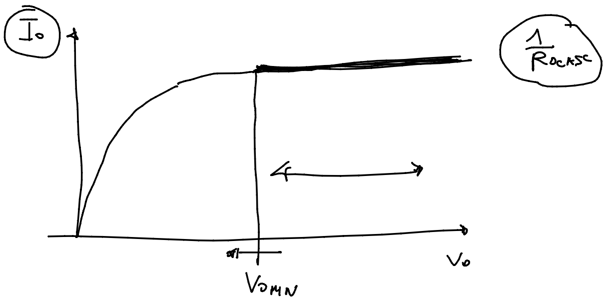
$$\frac{W}{L_4} = \frac{W}{L_2} = \frac{W}{L_1} \rightarrow V_{ov_2} = V_{ov_1} = V_{ov_4}$$

$$V_{G4} = 2V_T + 2V_{ov_1}$$

$$V_{GS_4} = V_T + V_{ov_4}$$

$$V_{DS_2} = V_T + V_{ov_1}$$

$$V_{OMIN} = \underbrace{V_T + \tilde{V}_{ov} + \tilde{V}_{ov}}$$



	CASCODE	CM
R_{OUT}	🙂	🙁
V_{OMIN}	🙁	🙂

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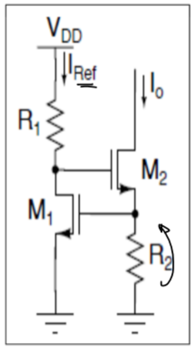
"V_t" standard

**



$$I_o = \frac{V_{GS1}}{R_2} = \frac{V_t + V_{OV1}}{R_2} = \frac{\cancel{V_t} + \sqrt{\frac{2I_{Ref}}{\underbrace{\mu C_{ox}(\frac{W}{L})_1}_{\beta_1}}}}{R_2}$$

$$\frac{\partial I_o}{\partial I_{REF}} = \frac{1}{R_2} \cdot \sqrt{\frac{2}{\beta_1}} \cdot \frac{1}{2} \cdot \frac{1}{(\sqrt{I_{REF}})} = \frac{1}{R_2 \beta_1 V_{o1}}$$

$$\sqrt{\frac{2 I_{REF}}{\beta_1}} = V_{ov_1} \Rightarrow \sqrt{I_{REF}} = \sqrt{\frac{\beta_1}{2}} \cdot V_{ov_1}$$

$$\boxed{S^{I_o}_{V_{DD}} = S^{I_o}_{I_{REF}} \cdot S^{I_{REF}}_{V_{DD}}}$$

$$S^{I_o}_{I_{REF}} = \frac{\partial I_o}{\partial I_{REF}} \cdot \frac{I_{REF}}{I_o} = \frac{1}{\cancel{R_2}\cancel{\beta_1}\cancel{V_{ov_1}}} \cdot \frac{\frac{\cancel{\beta_1}}{2} \cdot V_{ov_1}^{\cancel{2}}}{\frac{V_T + V_{ov_1}}{\cancel{R_2}}}$$

$$\boxed{S^{I_o}_{I_{REF}} = \frac{1}{2} \cdot \frac{V_{ov_1}}{V_T + V_{ov_1}}}$$

→ To find the sensitivity of the above circuit

i.e $$S^{I_o}_{V_{DD}} = S^{I_o}_{I_{REF}} \cdot S^{I_{REF}}_{V_{DD}}$$

This term has been derived above and its equal to

$$\boxed{S^{I_o}_{I_{REF}} = \frac{1}{2} \cdot \frac{V_{ov1}}{V_T + V_{ov1}}}$$

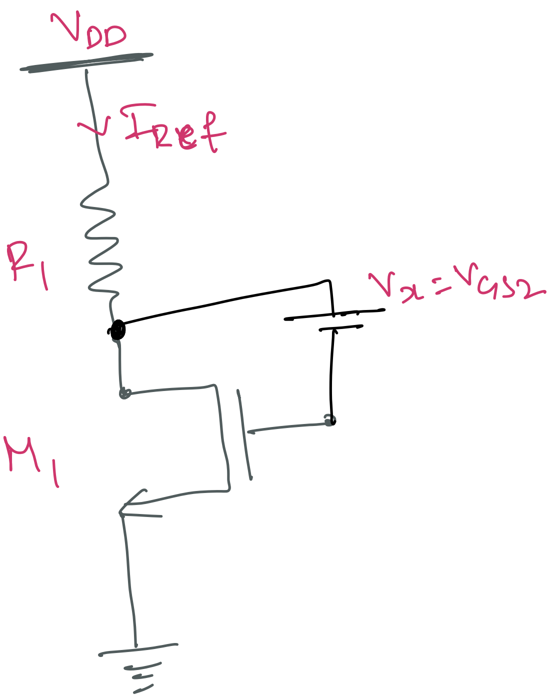
→ Now in order to find $S^{I_o}_{V_{DD}}$, we need to find $S^{I_{REF}}_{V_{DD}}$: sensitivity I_{REF} w.r.to V_{DD}

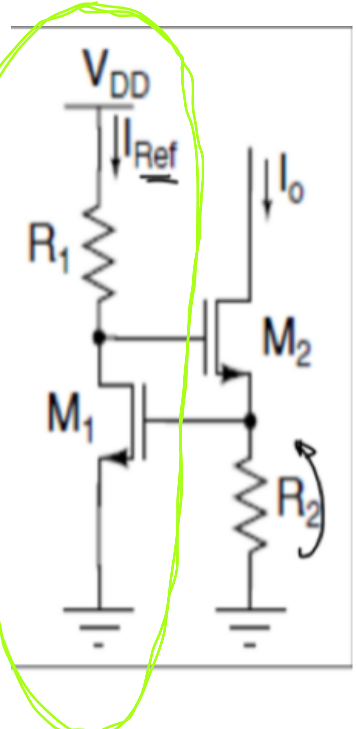
supply voltage

Now we will only look at left Branch of the circuit to find $S^{I_{REF}}_{V_{DD}}$

→ let the voltage applied b/w Drain & Gate of M_1 be V_x which is equal to V_{GS2}.

$\boxed{V_x = V_{GS2}}$ which is assumed to be constant





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∴ Equation for I_{REF} can be written as

$$I_{REF} = \frac{V_{DD} - V_{DS}}{R_1}$$

$$V_{DS_1} = V_{DG_1} + V_{GS_1}$$

$$= V_D - V_G + (V_G - V_S$$

$\Rightarrow V_D - V_G + V_G - V_S$

$V_{DG} = V_x$, $V_{GS_1} = V_{OV_1} + V_t$

$$\therefore \quad I_{REF} = \frac{V_{DD} - (V_x + V_{OV_1} + V_t)}{R_1}$$

$$\boxed{I_{REF} \Rightarrow \frac{V_{DD} - V_{OV_1} - V_t - V_x}{R_1}} \text{ — } a$$

So, if we assume V_{OV1} to be very small and doesn't have great impact on V_{DD}.

& V_t & V_x are independent of V_{DD}.

$\therefore$ we can say that

$$\boxed{\frac{\partial I_{REF}}{\partial V_{DD}} \simeq \frac{1}{R_1}}$$

To compute $S_{V_{DD}}^{I_{REF}} = \dfrac{V_{DD}}{I_{REF}} \dfrac{\partial I_{REF}}{\partial V_{DD}}$

we got this one

we need to compute it.

∴ From (a)

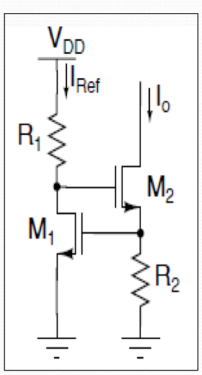
$$V_{DD} = I_{REF} R_1 + V_{OV_1} + V_t + V_x$$

$$\boxed{\frac{V_{DD}}{I_{REF}} = R_1 + \left(\frac{V_{OV_1} + V_t + V_x}{I_{REF}}\right) \simeq R_1}$$

If we suppose that these terms are negligible w.r.to voltage drop then we can approximate

$$\frac{V_{DD}}{I_{REF}} = R_1 \quad \therefore \quad S_{V_{DD}}^{I_{REF}} = 1$$

Sources Using Vt Standards



Can reduce sensitivity by making I_o depend on voltages other than V_{DD}, e.g., V_t.

$$I_o = \frac{V_{GS1}}{R_2} = \frac{V_t + V_{OV1}}{R_2} = \frac{V_t + \sqrt{\frac{2I_{Ref}}{\mu C_{ox}(\frac{W}{L})_1}}}{R_2}$$

If I_{Ref} is small and $(W/L)_1$ is sufficiently large $V_{OV} << V_t$: $\boxed{I_o \simeq \frac{V_t}{R_2}}$

Therefore, this circuit is known as a *threshold-referenced* bias circuit.

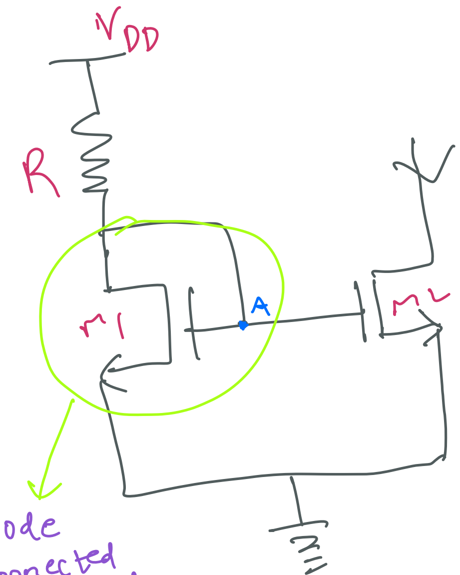
$$\begin{aligned} S_{V_{DD}}^{I_o} &= S_{I_{Ref}}^{I_o} \times S_{V_{DD}}^{I_{Ref}} \\ &\simeq \frac{I_{Ref}}{I_o} \times \frac{\partial I_o}{\partial I_{Ref}} \times \overbrace{S_{V_{DD}}^{I_{Ref}}}^{\simeq 1} \\ &\simeq \frac{1}{2}\frac{V_{OV1}}{V_t + V_{OV1}} \end{aligned}$$

For example, if $V_t = 1$ V, $V_{ov1} = 0.1$ V, and $S_{V_{DD}}^{I_{IN}} \simeq 1$

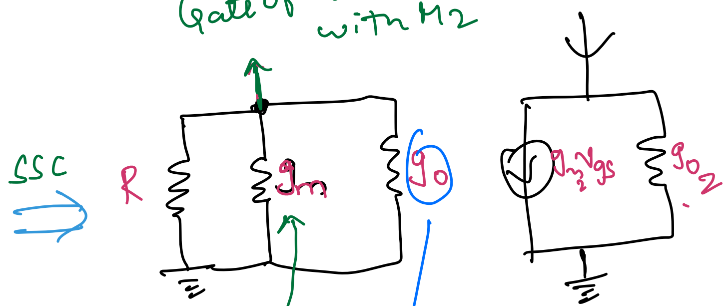
$$S_{V_{DD}}^{I_{OUT}} \simeq \frac{0.1}{2(1.1)} \simeq 0.045$$

→ It is also look at Current Generators in terms of Small Signal Circuit parameters

Let's look at common current Generator Circuit SSM



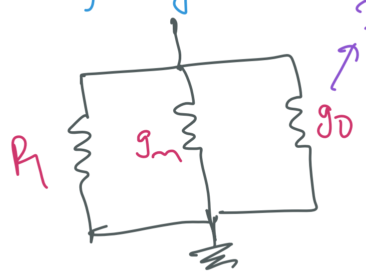
Gate of M2 is coincident with M2



Diode connected load for SSM

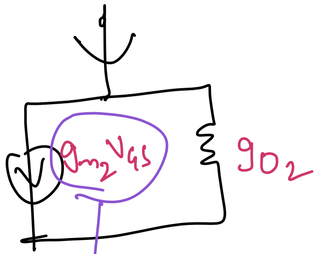
input impedence of M2 seen through the Node A

→ And there no Current flow between two Branches of the circuit because the link is connected to the Gate of M2 and the i/p impedence is very High



Intrinsic o/p conductance of M_1

∴ In SSM this circuit is stationary and do not exist i.e there are no variations in Gate voltages of the (M)

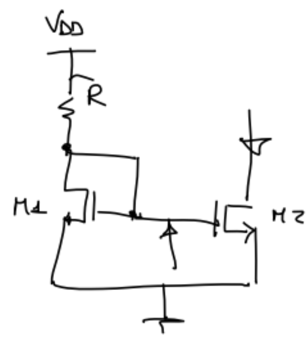


This also ZERO

∴ $v_{gs} = 0$

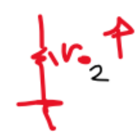
∴ In SSW the transconductance of M_2 can also be neglected and the only term we have is the "intrinsic o/p conductance" of M_2

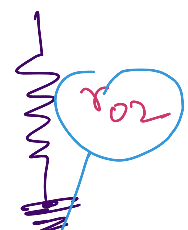
∴ In SSW



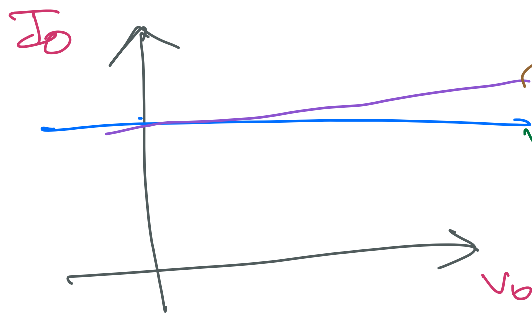
S.S.C →

The circuit is nothing but Resistance.





** The Higher is the value of r_{o2} it is that much Better. and the slope of the I_D & V_D graph is less if the Resistance is High.

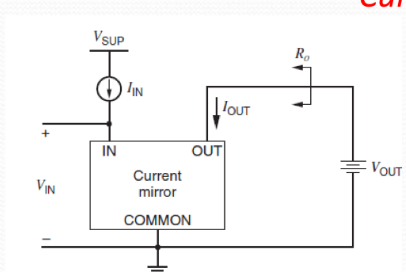


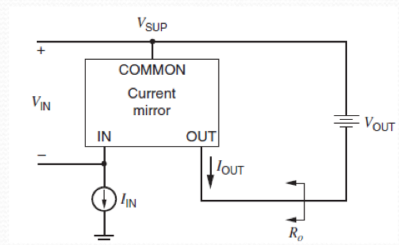
(with High r_{o2} the slope would be less)

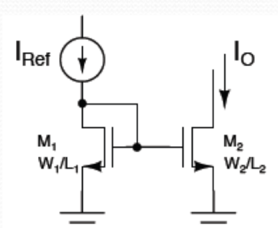
Current mirrors: Small signal analysis & Advanced schemes

* let's discuss the evolution of the concept of Current Generator in the form of Current Mirroring

Current Mirrors







$$\frac{I_O}{I_{Ref}} = \frac{\frac{1}{2}\mu C_{ox}\frac{W_2}{L_2}(V_{GS}-V_t)^2}{\frac{1}{2}\mu C_{ox}\frac{W_1}{L_1}(V_{GS}-V_t)^2} = \frac{\left(\frac{W_2}{L_2}\right)}{\left(\frac{W_1}{L_1}\right)} = K$$

We can scale I_{Ref} by modifying the dimensions of the device.

The equation suggests that scaling either or both widths and lengths can be used for this purpose.

Q Is there any advantage in either approach?

→ Here M_1 Transistor is always in saturation.
because $\underline{V_{DS} = V_{GS}}$

$$V_{DS} = V_{DG} + V_{GS}$$

∵ Drain and Gate are shorted the value of

$$\boxed{V_{DG} = 0}$$

→ M_2 is assumed to be in Saturation inorder to have Good insensitivity of the o/p Current w.r.to the o/p voltage. which is the Goal we want to achieve.

** Then we can take

$$\frac{I_O}{I_{Ref}} = \frac{\frac{1}{2}\mu C_{ox}\frac{W_2}{L_2}(V_{GS}-V_t)^2}{\frac{1}{2}\mu C_{ox}\frac{W_1}{L_1}(V_{GS}-V_t)^2} = \frac{\left(\frac{W_2}{L_2}\right)}{\left(\frac{W_1}{L_1}\right)} = K$$

current Gain that we have b/w I_O & I_{Ref}

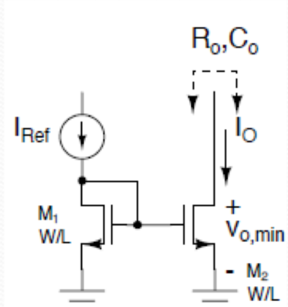
(or) we can say O/p current I_D is an exact duplicate of i/p current I_{Ref}.

Current Mirror Wish List

Ultimately, we want to use current mirrors to build current sources for biasing.

Ideally we want to
- Minimize the error in I_O
- Minimize $V_{o,min}$ (compliance voltage)
- Minimize C_o
- Maximize R_o

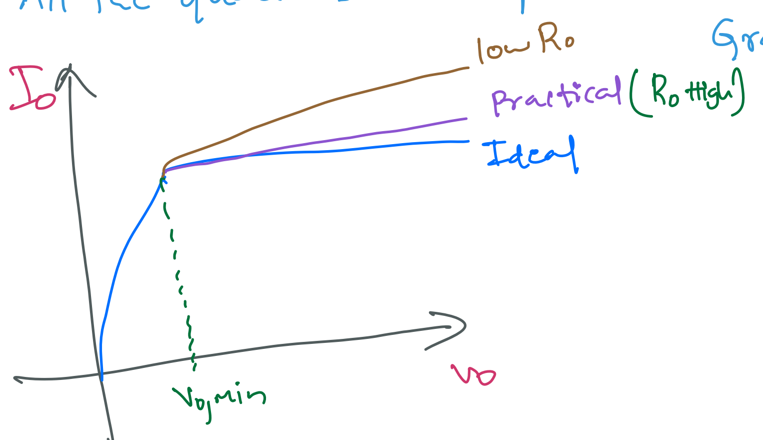
Minimum o/p voltage is related to overdrive voltage of M_2 in this case.



Two kinds of error:
- Systematic
  - $V_{DS2} \neq V_{DS1}$
  - IR drop in source connection
- Random
  - Mismatch in threshold voltages
  - Mismatch in $\beta = \mu C_{ox} W/L$

EK 8

All the qualities are represented in the Graph.



Because $\boxed{I_O = \frac{1}{R_O} V_O}$

Q Why this Ideal Behaviour characteristics are somehow problamatic ?

Ans First of all I_o is Rigorous duplicate of I_{Ref} ;

If V_{GS} of M_1 & M_2 are equal

Form factor is also equal &

V_{DS} is also equal in some sense because

$$\frac{I_o}{I_{Ref}} = \frac{\frac{1}{2}\mu_n C_{ox}\frac{W_2}{L_2}(V_{GS}-V_t)^2(1+\lambda V_{DS_2})}{\frac{1}{2}\mu_n C_{ox}\frac{W_1}{L_1}(V_{GS}-V_t)^2\underbrace{(1+\lambda V_{DS_1})}_{\text{Channel length Modulation}}}$$

*

Two kinds of error:

- Systematic
  - $V_{DS2} \neq V_{DS1}$
  - IR drop in source connection
- Random
  - Mismatch in threshold voltages V_t
  - Mismatch in $\beta = \mu C_{ox} W/L$

Doping ; Geometry etc may not be 100% same for both M_1 & M_2 .

∴ perfect mirroring behaviour may not occur.

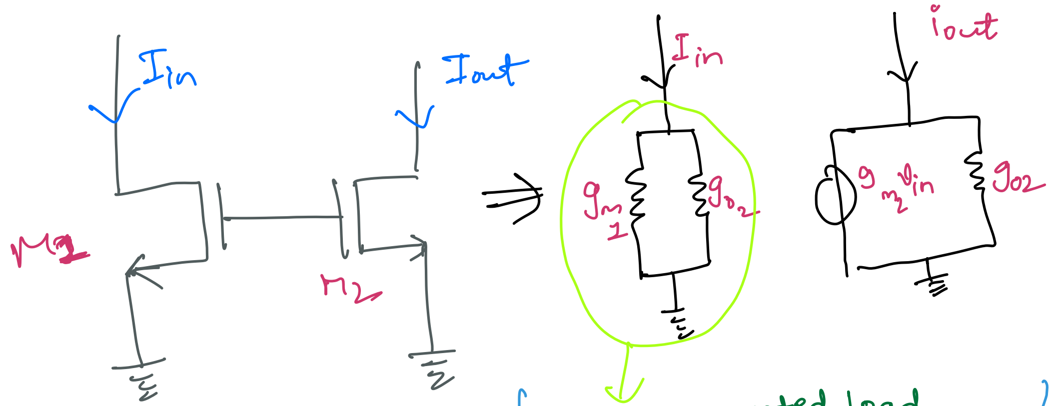
* BCD → power Devices

B → Bipolar

C → CMOS

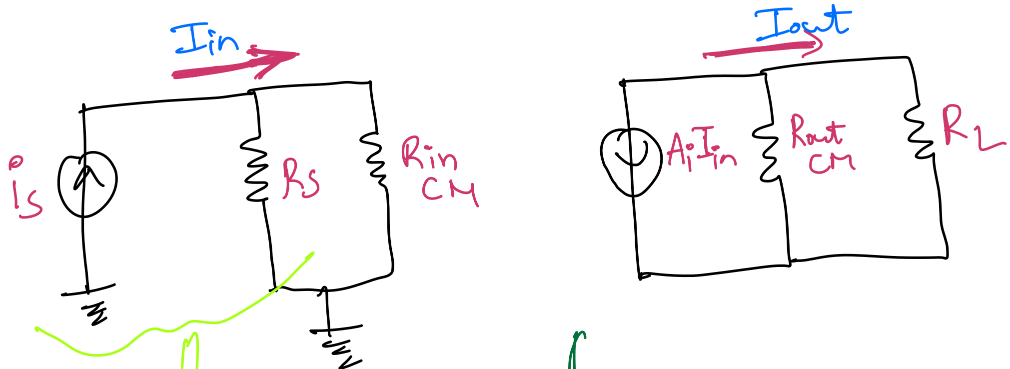
All these technologies can be manufactured on a chip and it is very expensive and quite Rare.

** To perform Small signal Analysis of current Mirror.



Diode connected load

The total SSM can be redrawn as

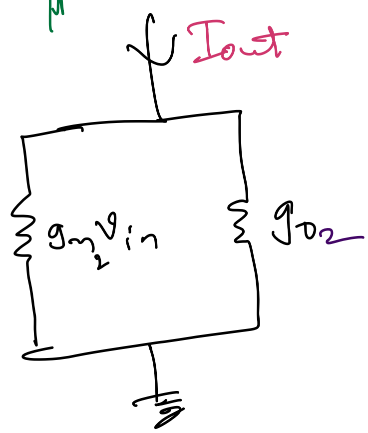


Current Source

with internal Resistance R_s which is supplying the input current I_{in}

we want R_s to be as high as possible for I_{in} to flow into Rin_{CM}

we want $Rout_{CM}$ to be as High as possible so that total I_{out} flows into R_L



$$\boxed{Rin_{CM} = \frac{1}{g_m + g_o}}$$

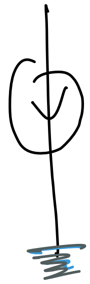
which is very less as Required

$$\boxed{Rout_{CM} = \frac{1}{g_{o2}}}$$

Is sufficiently High.

∴ This Topology is okay, we can do better than this if we use Cascode solution.

$$V_{in} \approx \frac{I_{in}}{g_{m1}}$$



$g_{m2} v_{in} = g_{m2} \dfrac{I_{in}}{g_{m1}}$

$$A_i = \left.\frac{i_o}{i_i}\right|_{v_i = 0} = \frac{g_{m2}}{g_{m1}} = 1$$

(if two Transistors are equal)

→ we will use Cascode Stage to Implement Current Generator.

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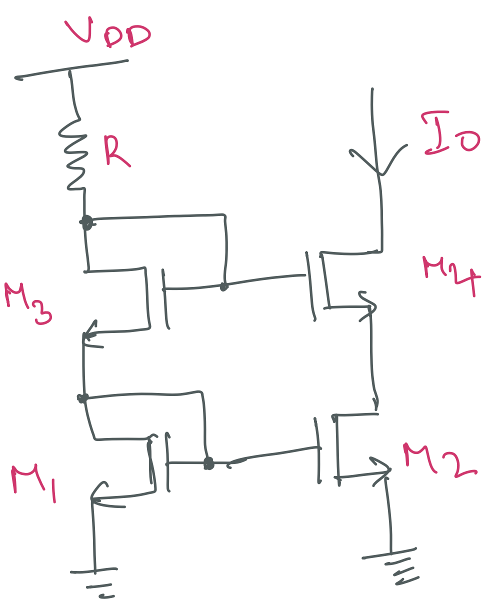
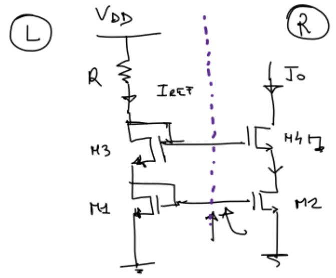


"SSA of the above Circuit:



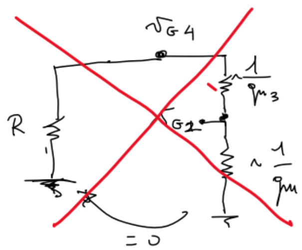
$I_{REF} = I_3 = I_1$

$I_o = I_4 = I_2$

If M_2 is SAT, $\frac{W}{L_1} = \frac{W}{L_2}$ $\Rightarrow I_2 \sim I_1 \Rightarrow I_o \sim I_{REF}$

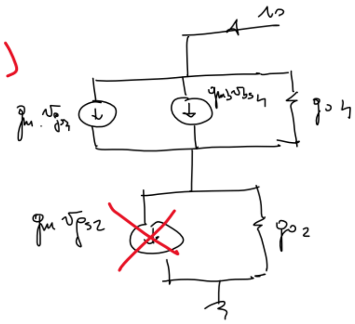
Very similar what we have done earlier in the class

S.S.C (L)



$v_{G4} = v_{G1} = 0$

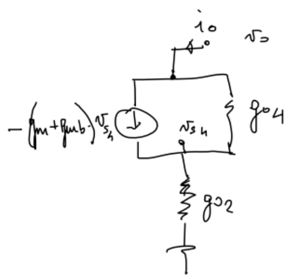
(R)



$v_{gs4} = \cancel{v_{g4}} - v_{s4}$

$v_{gs4} = -v_{s4}$

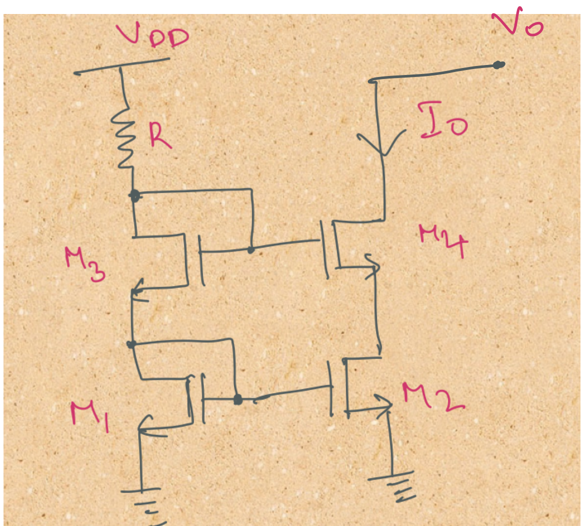
$v_{bs4} = -v_{s4}$



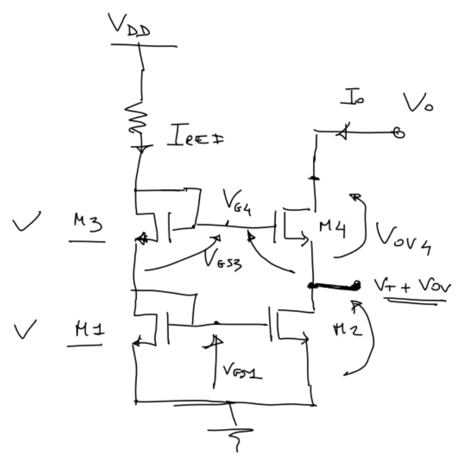
$$\rightsquigarrow R_{o_{casc}} = \frac{v_o}{i_o} = \frac{G_{T4}}{g_{o2} \cdot g_{o4}} \sim g_{m4} \cdot r_{o4} \cdot r_{o2}$$

$R_o =$  $R_{o_{casc}}$

**



large signal analysis to find out whether all Four Transistors are all four ... in saturation



$\boxed{V_{OMIN}}$

$V_{DS} = V_{GS} - V_T$

$V_{DS} = V_{GS}$

Wrong Answer

$[V_{DS_2} > V_{ov}]$

$V_{DS_2} = V_{ov_2}$

$\cancel{V_{OMIN} = V_{ov_2} + V_{ov_4}}$

$V_{GS1} = \underbrace{V_T + V_{ov_1}}$

$V_{GS3} = V_T + V_{ov_3}$

$\frac{W}{L_1} = \frac{W}{L_3} \rightarrow V_{ov_1} = V_{ov_3}$

$I_1 = I_3$

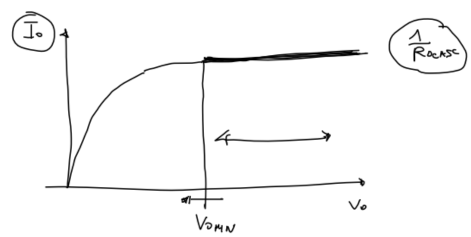
$\frac{W}{L_4} = \frac{W}{L_2} = \frac{W}{L_1} \rightarrow V_{ov_2} = V_{ov_1} = V_{ov_4}$

$V_{G4} = 2V_T + 2V_{ov_1}$

$V_{GS4} = V_T + V_{ov4}$

$V_{DS_2} = V_T + V_{ov_1}$

$V_{OMIN} = \underbrace{V_T + V_{ov} + V_{ov}}$



	CASCODE	CM
R_{out}	☺ ✓	☹ ✗
V_{OMIN}	☹ ✗	☺ ✓

⇒ Q What is the Minimum output voltage (V_{omin}) which allows all the 4 Transistors to be in saturation?

$\boxed{V_{omin}}$

Wrong Answer

Wrong Answer

$[V_{DS_4} > V_{OV}]$

$V_{DS_2} = V_{OV_2}$

$\cancel{V_{OMIN} = V_{OV_2} + V_{OV_4}}$

By

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